Introduction to Amazon EC2

SPL-200 - Version 1.3.19

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Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

Corrections, feedback, or other questions? Contact us at AWS Training and Certification.

Lab overview

This lab provides you with a basic overview of launching, resizing, managing, and monitoring an Amazon EC2 instance.

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers.

Amazon EC2's simple web service interface allows you to obtain and configure capacity with minimal friction. It provides you with complete control of your computing resources and lets you run on Amazon's proven computing environment. Amazon EC2 reduces the time required to obtain and boot new server instances to minutes, allowing you to quickly scale capacity, both up and down, as your computing requirements change.

Amazon EC2 changes the economics of computing by allowing you to pay only for capacity that you actually use. Amazon EC2 provides developers the tools to build failure resilient applications and isolate themselves from common failure scenarios.

Objectives

By the end of this lab, you should be able to do the following:

- Launch a web server with termination protection enabled.
- Monitor Your EC2 instance.
- **Modify** the security group that your web server is using to allow **HTTP** access.
- Resize your Amazon EC2 instance to scale.
- Test termination protection.
- Terminate your EC2 instance.

Start lab

1. To launch the lab, at the top of the page, choose Start Lab.

> **Caution:**
> You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose Open Console .

You are automatically signed in to the **AWS Management Console** in a new web browser tab.

> **Warning:**
> Do not change the Region unless instructed.

Common sign-in errors

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the Start Lab button from working as intended. If you experience an issue starting the lab:

Add the lab domain name to your pop-up or script blocker's allow list or turn it off.

Refresh the page and try again.

AWS services not used in this lab

AWS service capabilities used in this lab are limited to what the lab requires. Expect errors when accessing other services or performing actions beyond those provided in this lab guide.

Task 1: Launch Your Amazon EC2 Instance

In this task, you launch an Amazon EC2 instance with termination protection. **Termination protection** prevents you from accidentally terminating an EC2 instance. Your instance includes a User Data script that installs a simple web server.

3. At the top of the **AWS Management Console**, in the search bar, search for and choose EC2.

4. Choose **Launch instance** .

5. In the Name and tags section, enter **Web Server** in the Name box.

6. For **Amazon Machine Image (AMI)**, select an **Amazon Linux 2023 AMI**.

This lab only works with an **Amazon Linux 2023 AMI**.

> **Learn more:**
> An **Amazon Machine Image (AMI)** provides the information required to launch an instance, which is a virtual server in the cloud. An AMI includes:

- A template for the root volume for the instance (for example, an operating system or an application server with applications)
- Launch permissions that control which AWS accounts can use the AMI to launch instances
- A block device mapping that specifies the volumes to attach to the instance when it is launched

The Quick Start list contains the most commonly-used AMIs. You can also create your own AMI or select an AMI from the AWS Marketplace, an online store where you can sell or buy software that runs on AWS.

7. In the **Instance type** section, choose the **Instance type** drop down menu and choose **t3.micro**.

> **Learn more:**
> Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instance types comprise varying combinations of CPU, memory, storage, and networking capacity and give you the flexibility to choose the appropriate mix of resources for your applications. Each instance type includes one or more instance sizes, allowing you to scale your resources to the requirements of your target workload.

A **t3.micro** instance type has 2 virtual CPUs and 1 GiB of memory.

8. In the **Key pair (login)** section, locate the Key pair name - required drop down menu and choose **Proceed without a key pair (Not recommended)**.

> **Learn more:**
> Amazon EC2 uses public–key cryptography to encrypt and decrypt login information. To log in to your instance, you must create a key pair, specify the name of the key pair when you launch the instance, and provide the private key when you connect to the instance.

In this lab you do not log into your instance, so you do not require a key pair.

9. In the **Network settings** section, choose the **Edit** button. Make the following selections:

VPC - required: Choose the VPC with the name that contains **Lab VPC**

Subnet: Choose the Subnet with the name that contains **Public Subnet 1**

Learn more:

The Network indicates which Virtual Private Cloud (VPC) you wish to launch the instance into. You can have multiple networks, such as different ones for development, testing and production.

The **Lab VPC** was created using a CloudFormation template during the setup process of your lab. This VPC includes a public subnet and a security group.

10. In the **Firewall (security groups)** section, choose **Select existing security group**.

Common security groups: Enter **Web Server security group**.

Learn more:

A security group acts as a virtual firewall that controls the traffic for one or more instances. When you launch an instance, you associate one or more security groups with the instance. You add rules to each security group that allow traffic to or from its associated instances. You can modify the rules for a security group at any time; the new rules are automatically applied to all instances that are associated with the security group.

11. The **Configure storage** section default choices can be left alone.

Learn more:

Amazon EC2 stores data on a network-attached virtual disk called **Elastic Block Store**.

You launch the Amazon EC2 instance using a default 8 GiB disk volume. This is your root volume (also known as a 'boot' volume).

12. Expand the **Advanced details** section. Scroll down to the **Termination protection** drop down menu and set to **Enable**.

Learn more:

When an Amazon EC2 instance is no longer required, it can be terminated, which means that the instance is stopped and its resources are released. A terminated instance cannot be started again. If you want to prevent the instance from being accidentally terminated, you can enable termination protection for the instance, which prevents it from being terminated.

13. Scroll all the way to the bottom until you see a field for **User data - optional**.

Learn more:

When you launch an instance, you can pass user data to the instance that can be used to perform common automated configuration tasks and even run scripts after the instance starts.

Your instance is running Amazon Linux, so you provide a shell script that runs when the instance starts.

14. Copy the following text and paste it into the **User data - optional** field:

```
#!/bin/bash
dnf -y install httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.html
```

The script:

- Installs an Apache web server (httpd)
- Configures the web server to automatically start on boot
- Activates the Web server
- Creates a simple web page

15. Choose **Launch instance** .

Expected output:

✓ Success

Success

16. Choose **Instances** from the collapsible menu on the left pane. You may need to expand the menu to see this option.

The instance might appear in a pending state, which means it is being launched. It then changes to running, which indicates that the instance has started booting. When creating a new instance, there is usually a short time before you can access the instance.

17. Wait for your instance to display the following:

- **Instance state**: **Running** .
- **Status check**: **3/3 checks passed** .

Periodically refresh the page if you don't see a change in the **Instance state** or **Status check** values.

18. Select the checkbox next to your newly-created **Web Server** and choose the **Details tab** that displays detailed information about your instance.

Note:
To view more information in the **Details tab**, drag the window divider upwards.

Review the information displayed in the **Details tab**. It includes information about the instance type, security settings, network settings, and more. The instance receives a Public IPv4 DNS name that you can use to communicate with the instance from the Internet.

Task complete:
You have successfully launched your first Amazon EC2 instance.

Task 2: Monitor Your Instance

Monitoring is an important part of maintaining the reliability, availability, and performance of your Amazon Elastic Compute Cloud (Amazon EC2) instances and your AWS solutions.

19. Choose the **Status and alarms** tab.

Learn more:
With instance status monitoring, you can quickly determine whether Amazon EC2 has detected any problems that might prevent your instances from running applications. Amazon EC2 performs automated checks on every running EC2 instance to identify hardware and software issues.

Notice that the System status, Instance status and EBS status checks have passed.

20. Choose the **Monitoring tab**.

This tab displays CloudWatch metrics for your instance. Currently, there are not many metrics to display because the instance was recently launched.

You can choose a graph to see an expanded view.

Learn more:
Amazon EC2 sends metrics to Amazon CloudWatch for your EC2 instances. Basic (five-minute) monitoring is enabled by default. You can enable detailed (one-minute) monitoring.

21. Select the **Actions** menu (in the upper right of the console), choose **Monitor and troubleshoot** and select **Get system log**.

Expected output:

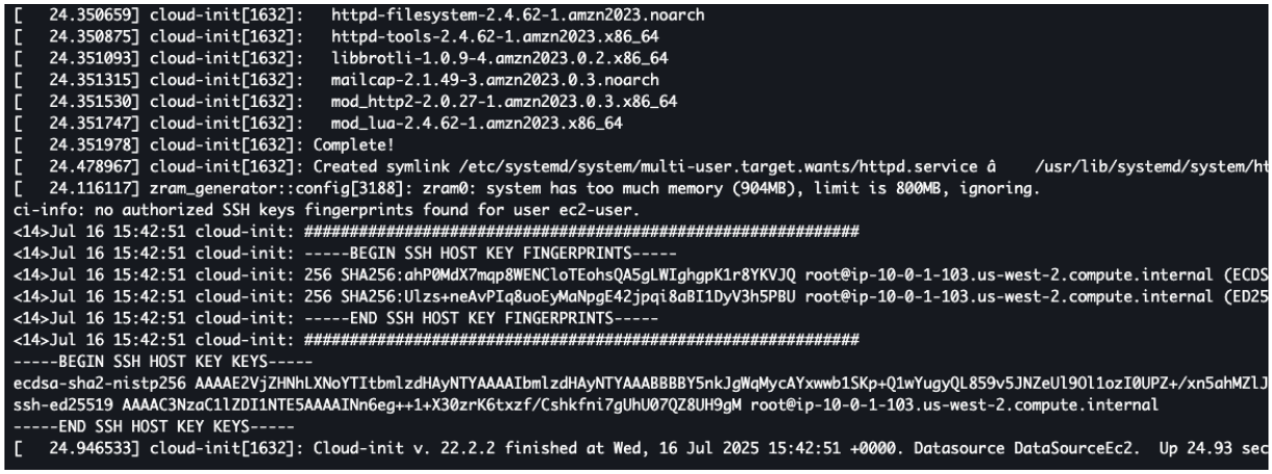


Image description: The preceding diagram depicts the system log console output for an Amazon EC2 instance.

Note:

If you do not see a system log, wait a few minutes and refresh the log screen until it appears.

The System Log displays the console output of the instance, which is a valuable tool for problem diagnosis. It is especially useful for troubleshooting kernel problems and service configuration issues that could cause an instance to terminate or become unreachable before its SSH daemon can be started.

22. Scroll through the output and note that the httpd package was installed from the user data that you added when you created the instance.

23. Scroll down to the bottom of the browser window and choose Cancel.

24. Select the checkbox next to the **Web Server**, choose the **Actions** menu, choose **Monitor and troubleshoot** and then choose **Get instance screenshot**.

Expected output:



Image description: The preceding diagram depicts what the Amazon EC2 instance console would look like if a screen were attached to it.

This shows you what your Amazon EC2 instance console would look like if a screen were attached to it.

Learn more:
If you are unable to reach your instance via SSH or RDP, you can capture a screenshot of your instance and view it as an image. This provides visibility as to the status of the instance, and allows for quicker troubleshooting.

25. Scroll down to the bottom of the browser window and choose Cancel.

Task complete:
You have explored several ways to monitor your instance.

Task 3: Update Your Security Group and Access the Web Server

When you launched the EC2 instance, you provided a script that installed a web server and created a simple web page. In this task, you access content from the web server.

26. Select the checkbox next to the **Web Server**, then choose the **Details tab**.

27. Copy the value of **Public IPv4 address** of your instance to your clipboard.

28. Open a new tab in your web browser, type http:// in the browser and paste the IP address you just copied, then press Enter.

Consider:
Are you able to access your web server? Why not?

You are not currently able to access your web server because the security group is not permitting inbound traffic on port 80, which is used for **HTTP** web requests. This is a demonstration of using a security group as a firewall to restrict the network traffic that is allowed in and out of an instance.

To correct this, you now update the security group to permit web traffic on port 80.

29. Keep the browser tab open, but return to the EC2 Management Console tab.

30. In the left navigation pane, select **Security Groups**.

31. Select the checkbox next to the Security group ID with the Security group name **Web Server security group**.

32. Choose the **Inbound rules** tab.

The security group currently has no rules.

33. Choose **Edit inbound rules**.

34. Choose **Add rule** then configure:

- **Type: HTTP**
- **Source: Anywhere-IPv4**

35. Choose **Save rules** .

The new Inbound **HTTP** rule creates an entry for both IPV4 IP address (0.0.0.0/0) as well as IPV6 IP address (::/0).

Note:
using “Anywhere”, or more specifically, using 0.0.0.0/0 or ::/0 is not a recommended best practice for production workloads.

36. Return to the web server tab that you previously opened and refresh the page.

Expected output:
You should see the message **Hello From Your Web Server!**

Task complete:

You have successfully modified your security group to permit **HTTP** traffic into your Amazon EC2 Instance.

Task 4: Resize Your Instance: Instance Type and EBS Volume

As your needs change, you might find that your instance is over-utilized (too small) or under-utilized (too large). If so, you can change the instance type. For example, if a **t3.micro** instance is too small for its workload, you can change it to an **t3.small** instance. Similarly, you can change the size of a disk.

Stop Your Instance

Before you can resize an instance, you must stop it.

Learn more:
When you stop an instance, it is shut down. There is no charge for a stopped EC2 instance, but the storage charge for attached Amazon EBS volumes remains.

37. On the EC2 Management Console, in the left navigation pane, choose **Instances**.

38. If it is not already selected, select the checkbox next to the **Web Server** instance.

39. Choose **Instance state** , then **Stop instance**.

40. Choose **Stop** .

Your instance performs a normal shutdown and then stops running. This may take a couple minutes.

41. Wait for the Instance State to display: **Stopped** .

Change The Instance Type

42. If it is not already selected, select the checkbox next to the **Web Server** instance.

43. Select the **Actions** menu, select **Instance settings** and **Change instance type**, then configure:

New instance type: t3.small

44. Choose **Change instance type** .

When the instance is started again it is a **t3.small**, which has twice as much memory as a **t3.micro** instance.

Resize the EBS Volume

45. In the left navigation pane, choose **Volumes** from the **Elastic Block Store** section.

46. Select the checkbox next to the listed volume.

47. In the **Actions** menu, choose **Modify volume**.

The disk volume currently has a size of 8 GiB. You now increase the size of this disk.

48. Change the **Size (GiB)** to: **10**

49. Choose **Modify** .

50. Choose **Modify** to confirm and increase the size of the volume.

Start the Resized Instance

You now start the instance again, which now has more memory and more disk space.

51. In left navigation pane, choose **Instances**.

52. Select the checkbox next to **Web Server**.

53. Choose **Instance state** and then choose **Start instance**.

Note:
An EBS volume being modified goes through a sequence of states: Modifying, Optimizing, and finally Complete.

Task complete:
You have successfully resized your Amazon EC2 Instance. In this task you changed your instance type from **t3.micro** to a **t3.small**. You also modified your root disk volume from 8 GiB to **10** GiB.

Task 5: Test Termination Protection

You can delete your instance when you no longer need it. This is referred to as terminating your instance. You cannot connect to or restart an instance after it has been terminated.

In this task, you learn how to use termination protection.

54. In left navigation pane, choose **Instances**.

55. Select the checkbox next to **Web Server**.

56. Choose **Instance state** and then choose **Terminate (delete) instance**.

57. Choose **Terminate (delete)** .

At this point you see the following error message on top of the page:

Failed to terminate (delete) an instance: The instance 'i-xxxxxxxx' may not be terminated. **Modify** its 'disableApiTermination' instance attribute and try again.

The above error is expected, and this is a safeguard to prevent the accidental termination of an instance. If you really want to terminate the instance, you need to disable the termination protection.

58. Choose **Actions** , choose **Instance settings** , and then choose **Change termination protection**.

59. Unselect the checkbox next to **Enable**.

60. Choose **Save** .

You can now terminate the instance.

61. Refresh the instance console screen.

62. Select the checkbox next to **Web Server**.

63. Choose **Instance state** , and then choose **Terminate (delete) instance** .

64. Choose **Terminate (delete)** .

Expected output:

The **Instance state** of the **Web Server** instance should change to Terminated after about 30 seconds. You may have to refresh the page a few times.

Task complete:
You have successfully tested termination protection and terminated your instance.

Knowledge check

This knowledge check is designed to gauge your understanding of the material covered.

Why this matters: Completing this knowledge check will help reinforce your grasp of the content and highlight any areas that may need further review.

Launch knowledge check ■

Conclusion

You successfully did the following:

- Launched a web server with termination protection enabled.
- Monitored Your EC2 instance.
- Modified the security group that your web server is using to allow **HTTP** access.
- Resized your Amazon EC2 instance to scale.
- Tested termination protection.
- Terminated your EC2 instance.

End lab

Follow these steps to close the console and end your lab.

65. Return to the **AWS Management Console**.

66. At the upper-right corner of the page, choose **AWSLabsUser**, and then choose **Sign out**.

67. Choose **End Lab** and then confirm that you want to end your lab.